Architectural Issues
— Feedforward
(控制结构问题 — 前馈设计)

Feedforward for General Reference Tracking

- **CL System Output due to Reference Inputs**(由参考输入引起的闭环系统输出)

Ideally, perfect tracking can be achieved even during the transient if one can choose feedforward TF $F(s)$ such that (理想情况下，通过选择前馈传递函数$F(s)$，即使在瞬态过程中也可以实现较好的跟踪)

$$Y(s) = T(s)F(s)R(s)$$

$$T(s)F(s) = 1 \quad \text{or} \quad F(s) = 1/T(s)$$

In reality, the above exact inversion may not be done due to(实际情况中，可能不允许采用上述精确求逆的方法，由于):

(1) The CLTF $T(s)$ has unstable CL zeros (闭环传递函数有不稳定的闭环零点).

(2) The inversion $1/T(s)$ may not be proper and future reference input trajectory is not known, which means $\bar{R}(s) = R(s)/T(s)$ is not calculable or implementable.($T(s)$的逆可能不是正则的，并且将来的参考输入轨迹未知，这意味着$\bar{R}(s) = R(s)/T(s)$ 无法计算或可实施)

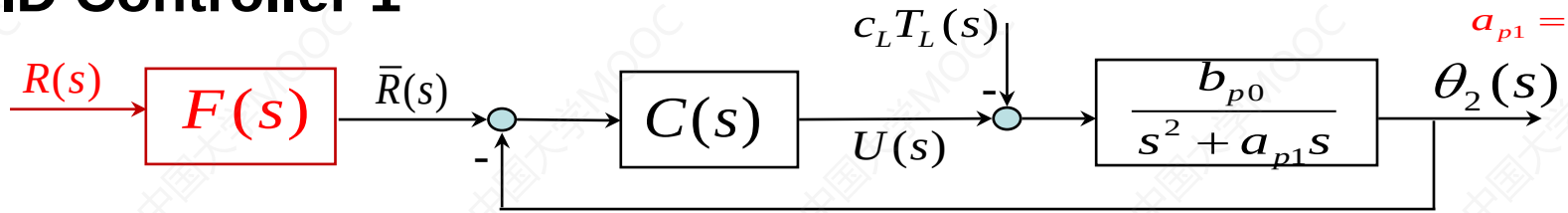
In these cases, instead of the above exact inversion, approximated inversions may be carried out. For example, in the second case, fast poles may be inserted to make the resulting TF proper:(在这些情况下，可以用近似求逆代替上述精确求逆。例如，在第二种情况下，可增加快速极点以使得到的传递函数是正则的)

$$F(s) = \frac{1}{T(s)} \frac{1}{(\tau_h s + 1)^k}, \quad \tau_h << 1$$

PID Control of Rigid Belt ECP Emulator

- **PID Controller 1**

$b_{p0} = 623$
$a_{p1} = 3.5$



A proper PID controller has been previously synthesized to place CL poles at (之前设计了一个正则PID控制器，其放置闭环极点在)

$$p_{CL1,2d} = -20 \pm 20j, \qquad p_{CL3,4d} = -40$$

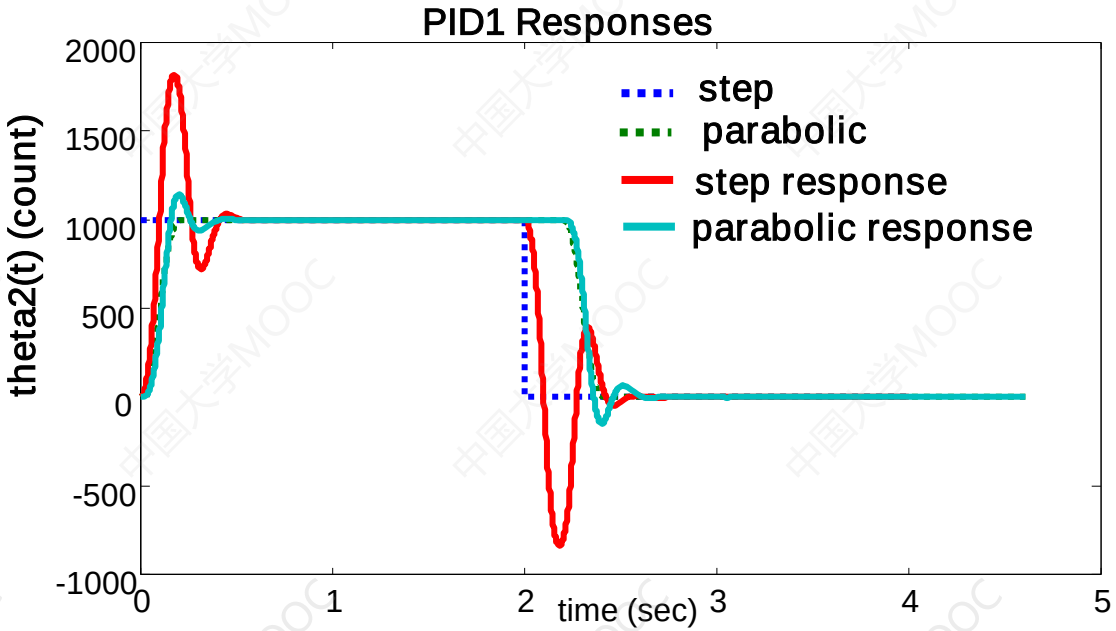
$$C(s) = \frac{8.33s^2 + 205s + 2054}{s^2 + 117s}$$

$$T_n(s) = \frac{PC}{1+PC} = \frac{N_p N_c}{A_{CL}(s)} = \frac{b_{p0}\left(8.33s^2 + 205s + 2054\right)}{\left((s+20)^2 + 20^2\right)(s+40)^2}$$

$$Y(s) = T(s)F(s)R(s)$$

PID Control Experimental Results

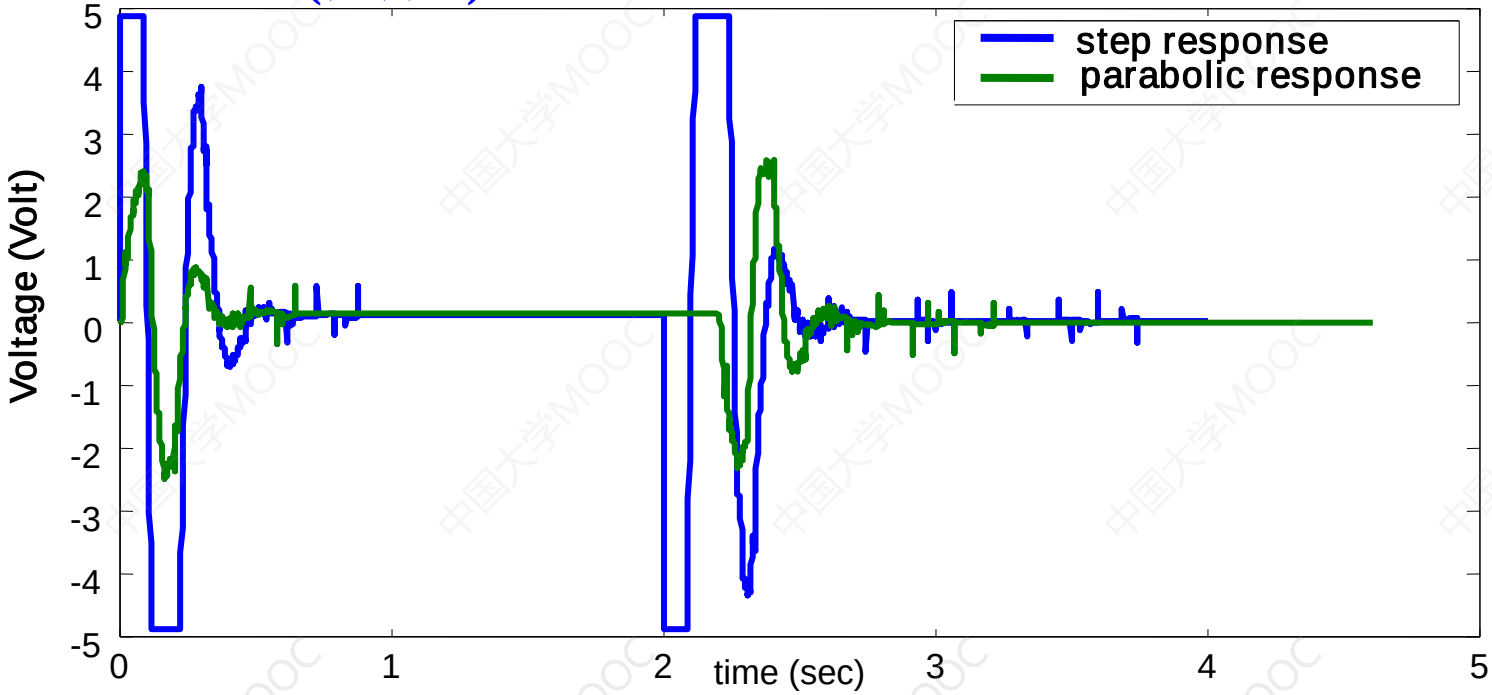
- **Trajectory Planning for Reducing/Eliminating Actuator Saturation**
  (减少/消除执行器饱和的轨迹规划)



The above figure shows the improved transient response of using trajectory planning (less overshoot and no degradation of settling time). **(上图可看出，使用轨迹规划后瞬态响应得到了改善(超调量减少，调整时间没有降低))**

PID Control Experimental Results

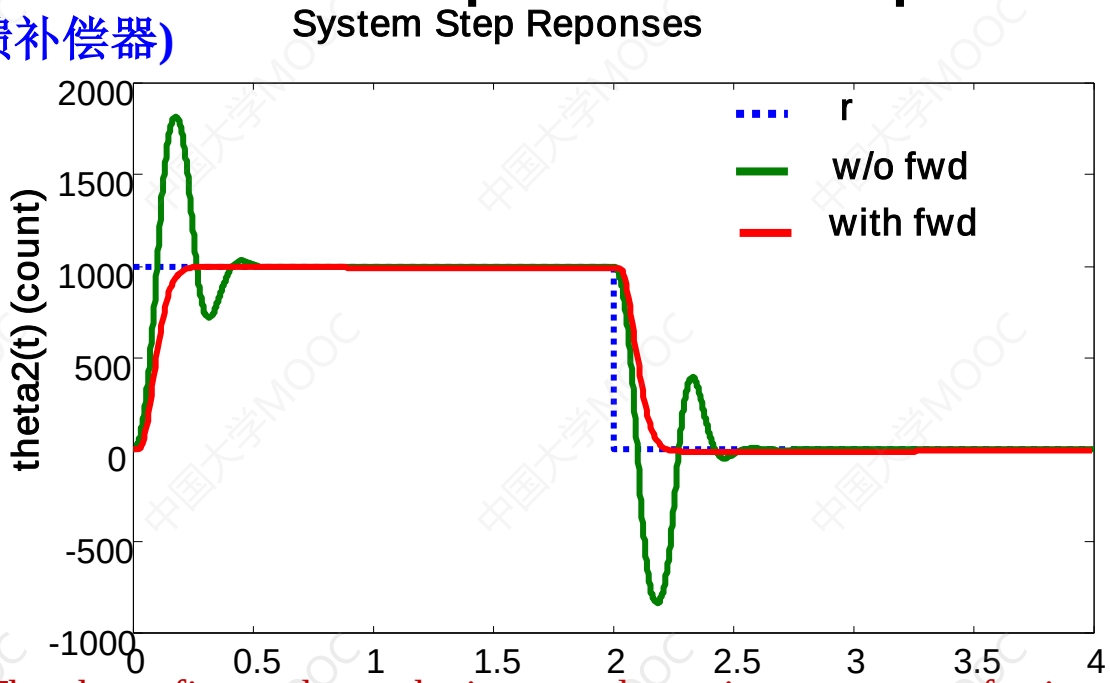
- **Control Effort (控制量)**



The above figure shows the efficacy of trajectory planning in reducing/eliminating actuator saturation (上图显示了轨迹规划在减小/消除执行器饱和方面的效果).

PID Control Experimental Results

- **Feedforward Compensator for Improved Transient (改进瞬态的前馈补偿器)**



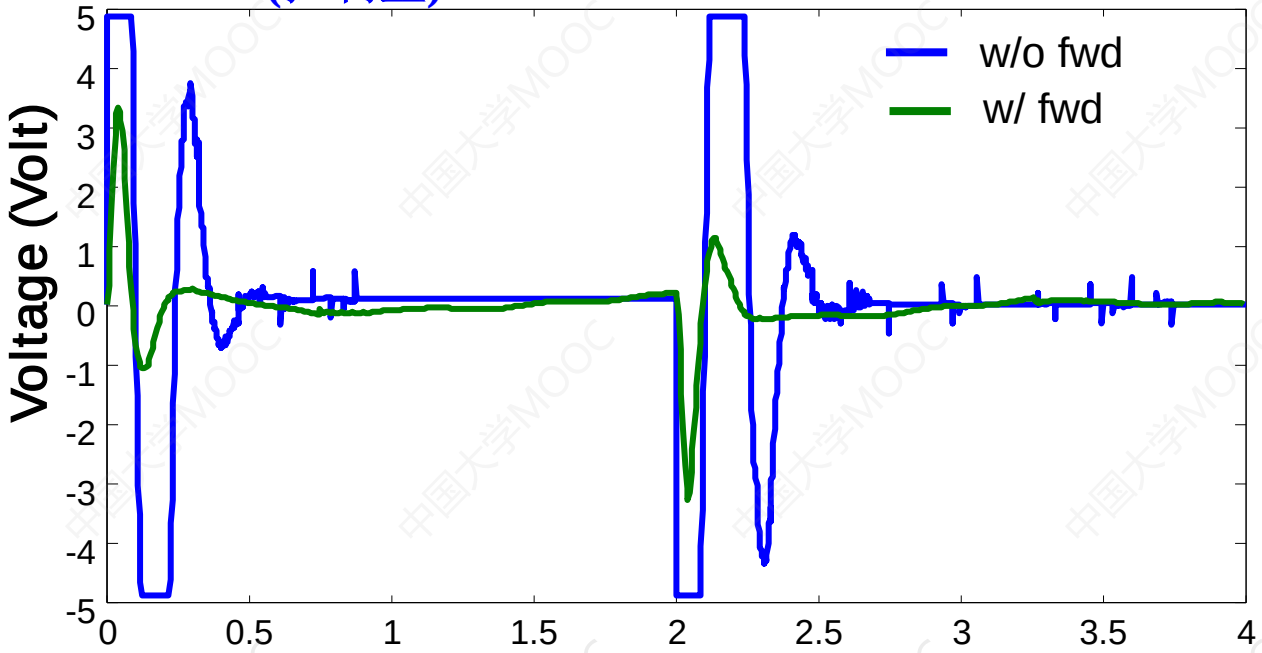
The comparison is done between the CL system with and without $F(s)$ in canceling the stable zeros of CLTF $T(s)$(比较了有和没有$F(s)$的闭环系统对消去闭环传函 $T(s)$稳定零点的效果)

$$F(s) = \frac{246}{s^2 + 24.6s + 246}$$

The above figure shows the improved transient response of using simple feedforward compensation (no overshoot vs huge overshoot) (从上图可以看出，采用简单的前馈补偿可以改善瞬态响应(无超调对比大超调)).

PID Control Experimental Results

- **Control Effort (控制量)**

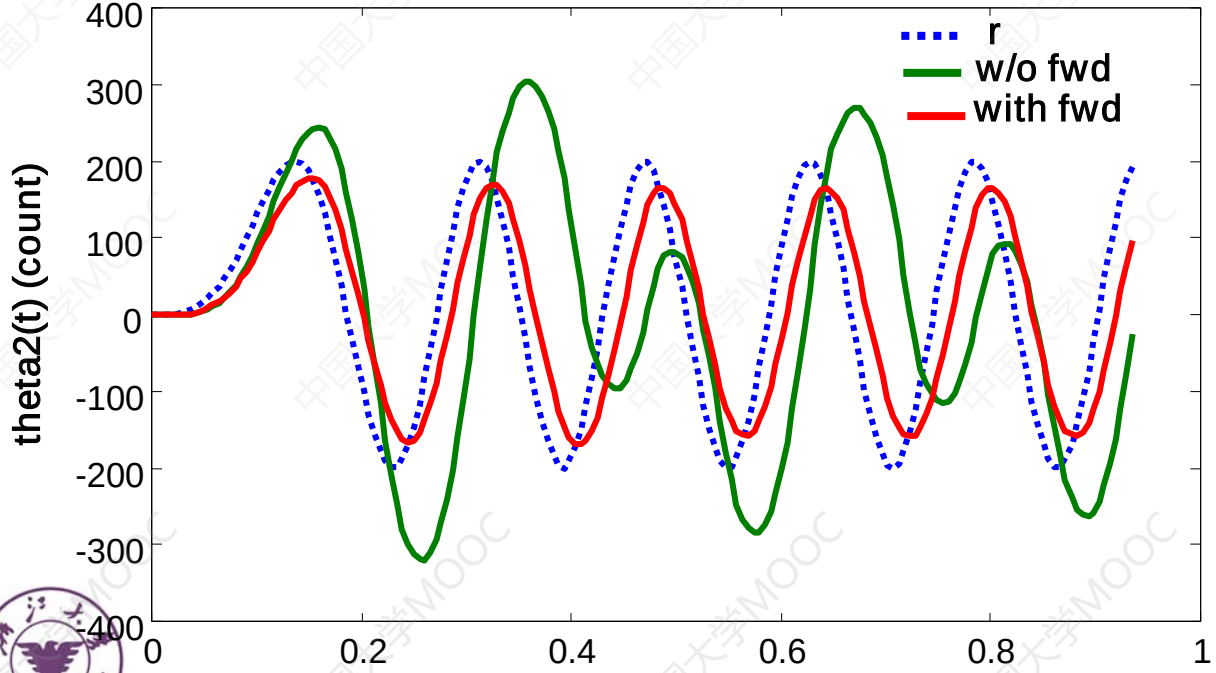


The above figure shows the efficacy of feedforward in reducing or eliminating actuator saturation due to improved transient performance (从上图可看出，由于改善了瞬态性能，前馈可以有效地减小或消除执行器饱和).

PID Control Experimental Results

- **Feedforward Compensator for Improved Reference Tracking**
  (改进参考跟踪的前馈补偿器)

Tracking a sinusoidal input of $r(t)=200\sin(40t)$ counts **(跟踪一个正弦参考输入)**



The comparison is done between the CL system with and without $F(s)$ in canceling the stable zeros of CLTF $T(s)$ **(比较了有和没有$F(s)$的闭环系统对消去闭环传函$T(s)$稳定零点的效果)**

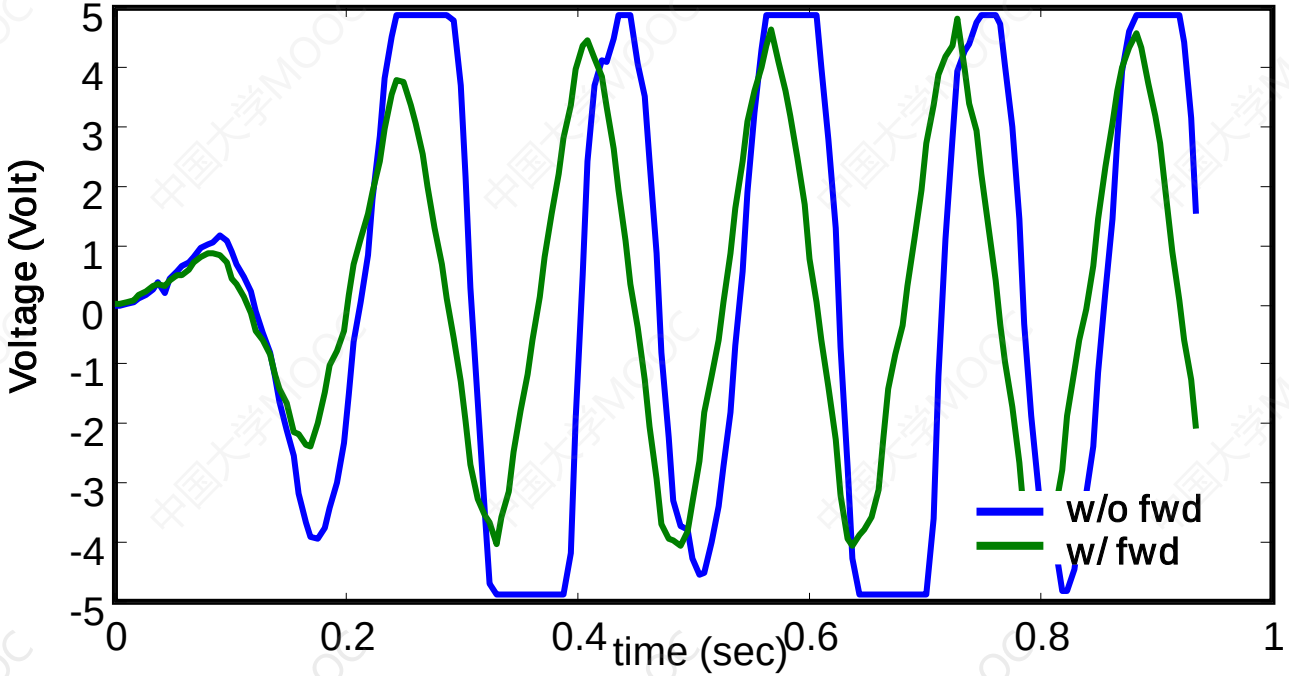
$$F(s)=\frac{1}{T(s)}\frac{1}{(\tau_h s+1)^4}$$

$$=\frac{A_d^c(s)}{8.33b_{p0}\left(s^2+24.6s+246\right)}\frac{1}{(\tau_h s+1)^4}$$

$$\tau_h=0.01$$

PID Control Experimental Results

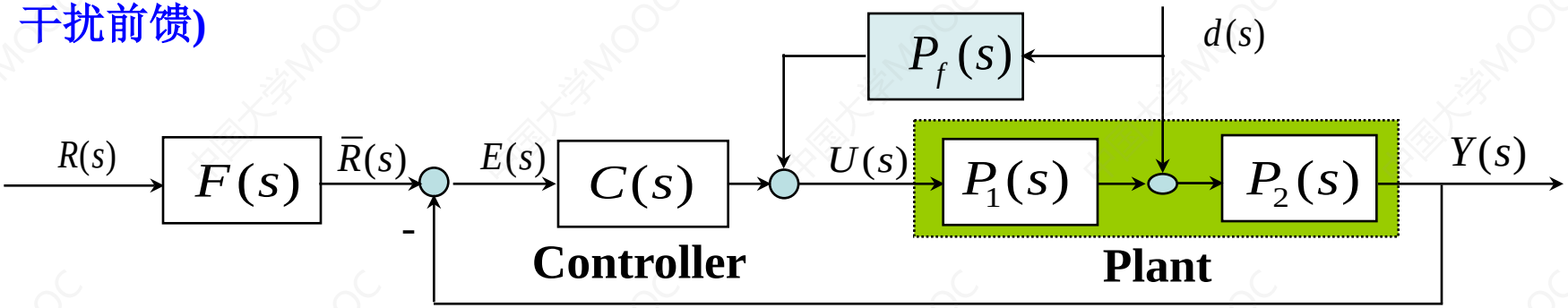
- **Control Effort** (控制量)



The above figure shows the efficacy of feedforward in reducing or eliminating actuator saturation due to improved transient performance (从上图可看出，由于改善了瞬态性能，前馈可以有效地减小或消除执行器饱和).

Feedforward for Disturbance Rejection

- **Disturbance Feedforward for *Measured Disturbance*** (测量干扰的干扰前馈)



- **Output due to Disturbance** (因干扰引起的输出)

$$Y_d(s) = S(s)P_2(s)\left[1 + P_1(s)P_f(s)\right]d(s)$$

- **Ideal Disturbance Feedforward TF** (理想干扰前馈传递函数)

$$P_1(s)P_f(s) = -1 \Rightarrow P_f(s) = -\frac{1}{P_1(s)}$$

Summary

- **Feedforward (前馈)**
  - Feedforward for Reference Tracking（参考跟踪的前馈）
  - Feedforward for Disturbance Rejection（干扰抑制的前馈）